

Report

Atticus Tarleton

2026-06-08

The below code sets the project root to the right location, making file paths easier to manage

```
import os
from pathlib import Path

_project_root = Path.cwd()
while not (_project_root / ".git").exists():
    _project_root = _project_root.parent
    if _project_root == _project_root.parent:
        raise RuntimeError(".git not found - are you inside the repo?")
os.chdir(_project_root)
print(os.getcwd())
```

/Users/AGT/Desktop/ORCA/OkemoDataLearningProject

Creating the database and some more data engineering

The below code block is going to create a database connection to the large and small clean healthcare databases made in the engineering.qmd file

```
import duckdb

con = duckdb.connect()
con.execute(f"""
    CREATE OR REPLACE TABLE small_healthcare AS
    SELECT * FROM read_parquet('{ "Data/clean_data/small_2024_health_data.parquet" }')
""")
con.execute(f"""
    CREATE OR REPLACE TABLE large_healthcare AS
```

```

SELECT * FROM read_parquet('{"Data/clean_data/large_2024_health_data.parquet"}')
""")
con.execute(f"""
CREATE OR REPLACE TABLE small_combined_healthcare AS
SELECT * FROM read_parquet('{"Data/clean_data/small_2024_health_data.parquet"}')
WHERE st_abbrev = 'VT' OR st_abbrev = 'US'
""")

```

<_duckdb.DuckDBPyConnection at 0x107f668f0>

Questions to answer 1) How does Vermont's ratios of health care professions look compared to the US as a whole - graph 1: Bar plot of ratio of vermont next to US ratio - will need to do some pivoting of data and such - graph 2: do the same as above, but instead by summing the northeast region - how does vermont compare to the local region - questions: - include vermont or not? - take the ratios of each state and average them, or sum the states values and average them? 2) States with the most physicians, and their total population - basically, do the numbers align - and do the same with the least physicians 3) States with the greatest ratio of physicians, and with the smallest ratio

This code chunk below sets up the datasets that will be plotted and used in the future

```

import pandas as pd

small_health_df = pd.read_parquet('Data/clean_data/small_2024_health_data.parquet')
large_health_df = pd.read_parquet('Data/clean_data/large_2024_health_data.parquet')

occupation_count_columns = small_health_df.columns[2:35].tolist()

small_ratio_df = small_health_df.copy()
for i in occupation_count_columns:
    small_ratio_df[i] = small_ratio_df[i]/small_ratio_df['popn_pums_23']

small_combined_ratio_df = small_ratio_df[(small_ratio_df['st_abbrev'] == 'VT') | (small_ratio_df['region'] == 'NE')]

small_combined_ratio_melted = small_combined_ratio_df.melt(id_vars='st_abbrev',
    value_vars=occupation_count_columns,
    var_name='occupation', value_name='ratio')

# to find columns with nan values so they can be removed
with pd.option_context('display.max_columns', None):
    print(small_health_df[(small_health_df['st_abbrev'] != 'VT') & (small_health_df['region'] != 'NE')])
    #columns with nan values: podtrst, chiro, opto, dietn, pharm_aide, med_transcrp, vetin_a

```

```

occupation_count_columns_ne = occupation_count_columns
occupation_count_columns_ne.remove('podtrst_23')
occupation_count_columns_ne.remove('chiro_23')
occupation_count_columns_ne.remove('opto_23')
occupation_count_columns_ne.remove('dietn_23')
occupation_count_columns_ne.remove('pharm_aide_23')
occupation_count_columns_ne.remove('med_transcrp_23')
occupation_count_columns_ne.remove('vetin_asst_23')
occupation_count_columns_ne.remove('phleb_23')

ne_without_vt_sum_df = small_health_df[(small_health_df['st_abbrev'] != 'VT') & (small_health_df['fips_st'] != 'VT')]
vt_only_df = small_health_df.iloc[45]
ne_without_vt_sum_df['st_abbrev'] = 'NE'
ne_without_vt_sum_df['region'] = 'NE'
ne_without_vt_sum_df['fips_st'] = '-99' #representing a null value
vt_and_ne_df = pd.DataFrame([vt_only_df, ne_without_vt_sum_df])
vt_and_ne_df = vt_and_ne_df.reset_index().drop('index', axis=1)

vt_and_ne_ratio_df = vt_and_ne_df.copy()
for i in occupation_count_columns:
    vt_and_ne_ratio_df[i] = vt_and_ne_ratio_df[i]/vt_and_ne_ratio_df['popn_pums_23']

vt_and_ne_ratio_melted = vt_and_ne_ratio_df.melt(id_vars='st_abbrev',
        value_vars=occupation_count_columns,
        var_name='occupation', value_name='ratio')

```

	fips_st	st_abbrev	phys_wkforc_23	pa_23	rn_23	dent_23	pharm_23	\
6	9	CT	14223	2646.0	38042	2230.0	3519.0	
19	23	ME	4303	1153.0	15819	719.0	1603.0	
21	25	MA	33466	5041.0	88714	5602.0	8215.0	
29	33	NH	4448	776.0	16353	1020.0	1263.0	
39	44	RI	4325	519.0	11593	464.0	1444.0	

	vetin_23	podtrst_23	chiro_23	opto_23	psychol_23	conslrs_23	socwk_23	\
6	1415.0	NaN	916.0	422.0	5317.0	5200	7929	
19	575.0	NaN	NaN	NaN	1097.0	2519	3494	
21	2926.0	NaN	1228.0	939.0	10273.0	17436	17519	
29	716.0	NaN	NaN	NaN	1218.0	2635	1917	
39	360.0	NaN	NaN	NaN	1385.0	2275	2866	

	pt_23	ot_23	resp_ther_23	spech_path_23	massg_ther_23	dietn_23	\
--	-------	-------	--------------	---------------	---------------	----------	---

6	4222	1835.0	1124.0	2328.0	2204.0	1168.0
19	1208	1112.0	467.0	927.0	852.0	630.0
21	9269	5265.0	1941.0	5208.0	4560.0	3268.0
29	2314	1883.0	575.0	1154.0	1273.0	NaN
39	1306	540.0	332.0	595.0	570.0	306.0

	medmgr_23	med_secrtrs_23	clin_lab_techs_23	diag_reltd_techs_23	\
6	11075	6857	3736	6002	
19	4103	2894	1524	2265	
21	23786	15580	9987	10529	
29	4293	2353	1007	1934	
39	3614	3013	1990	1482	

	emt_parmdcs_23	hlth_sup_techs_23	med_rec_spec_23	med_asst_23	\
6	3124	9597	2119.0	7150.0	
19	1060	4413	794.0	3023.0	
21	6591	19000	2969.0	13475.0	
29	689	4226	730.0	2292.0	
39	816	2739	443.0	2049.0	

	pharm_aide_23	med_transcrp_23	vetin_asst_23	phleb_23	oth_hlthcar_23	\
6	NaN	213.0	725.0	1608.0	2790	
19	NaN	NaN	NaN	NaN	974	
21	905.0	616.0	1379.0	3628.0	4664	
29	NaN	NaN	449.0	389.0	1132	
39	NaN	NaN	NaN	669.0	961	

	pers_care_aide_23	nurse_aide_23	popn_pums_23	popn_ge60_23	popn_24	\
6	18009	27475	3598348	907442	3675069	
19	7496	11300	1377400	411816	1405012	
21	34794	55493	6992395	1697264	7136171	
29	3932	8974	1387835	381063	1409032	
39	3658	9804	1095371	279082	1112308	

	region
6	NE
19	NE
21	NE
29	NE
39	NE

Exploratory Data Analysis

The code chunk below answers the question of which states have the most physicians, and which ones have the least using Pandas

```
physician_sorted_df = small_health_df.drop(51).sort_values(by='phys_wkforc_23', ascending=False)
print("\n the 10 states with the most physicians")
display(physician_sorted_df.head(10))
print("\n the 10 states with the fewest physicians")
display(physician_sorted_df.tail(10))
```

the 10 states with the most physicians

	st_abbrev	phys_wkforc_23	popn_pums_23
4	CA	115435	39242785
32	NY	88345	19872319
43	TX	73675	29640343
9	FL	64392	21928881
38	PA	47128	12986518
35	OH	39774	11780046
13	IL	38946	12692653
21	MA	33466	6992395
22	MI	31279	10051595
33	NC	30755	10584340

the 10 states with the fewest physicians

	st_abbrev	phys_wkforc_23	popn_pums_23
39	RI	4325	1095371
19	ME	4303	1377400
12	ID	3814	1893297
7	DE	3375	1005873
26	MT	3259	1105072
41	SD	2810	899194
45	VT	2354	645254
34	ND	2244	779361

	st_abbrev	phys_wkforc_23	popn_pums_23
1	AK	2158	733971
50	WY	960	579761

The code chunk below answers the of which states have the most physicians, and which ones have the least using SQL

```
print("\n the 10 states with the most physicians")
display(con.execute("""SELECT st_abbrev, phys_wkforc_23, popn_pums_23
    FROM small_healthcare WHERE st_abbrev != 'US'
    ORDER BY phys_wkforc_23 DESC LIMIT 10""").df())

print("\n the 10 states with the fewest physicians")
display(con.execute("""SELECT st_abbrev, phys_wkforc_23, popn_pums_23
    FROM small_healthcare WHERE st_abbrev != 'US'
    ORDER BY phys_wkforc_23 LIMIT 10""").df())
```

the 10 states with the most physicians

	st_abbrev	phys_wkforc_23	popn_pums_23
0	CA	115435	39242785
1	NY	88345	19872319
2	TX	73675	29640343
3	FL	64392	21928881
4	PA	47128	12986518
5	OH	39774	11780046
6	IL	38946	12692653
7	MA	33466	6992395
8	MI	31279	10051595
9	NC	30755	10584340

the 10 states with the fewest physicians

	st_abbrev	phys_wkforc_23	popn_pums_23
0	WY	960	579761
1	AK	2158	733971
2	ND	2244	779361
3	VT	2354	645254
4	SD	2810	899194
5	MT	3259	1105072
6	DE	3375	1005873
7	ID	3814	1893297
8	ME	4303	1377400
9	RI	4325	1095371

The code chunk below answers the question of what states have the largest and smallest ratio of health care professionals using Pandas

```
small_ratio_df.sort_values(by='phys_wkforc_23', ascending=False)[['st_abbrev', 'phys_wkforc_23', 'popn_pums_23']]
```

	st_abbrev	phys_wkforc_23	popn_pums_23
8	DC	0.009578	672079
21	MA	0.004786	6992395
32	NY	0.004446	19872319
11	HI	0.003972	1445635
6	CT	0.003953	3598348
39	RI	0.003948	1095371
20	MD	0.003654	6170738
45	VT	0.003648	645254
38	PA	0.003629	12986518
37	OR	0.003590	4238714
35	OH	0.003376	11780046
7	DE	0.003355	1005873
5	CO	0.003228	5810774
30	NJ	0.003208	9267014
29	NH	0.003205	1387835
47	WA	0.003170	7740984
41	SD	0.003125	899194
19	ME	0.003124	1377400
22	MI	0.003112	10051595
13	IL	0.003068	12692653
51	US	0.003039	332387543
48	WV	0.003028	1784462

	st_abbrev	phys_wkforc_23	popn_pums_23
23	MN	0.002981	5713716
26	MT	0.002949	1105072
4	CA	0.002942	39242785
1	AK	0.002940	733971
9	FL	0.002936	21928881
42	TN	0.002913	6986082
46	VA	0.002912	8657499
33	NC	0.002906	10584340
34	ND	0.002879	779361
25	MO	0.002864	6168181
40	SC	0.002779	5212774
27	NE	0.002728	1965926
44	UT	0.002658	3331187
15	IA	0.002657	3195937
18	LA	0.002639	4621025
36	OK	0.002629	3995260
49	WI	0.002617	5892023
31	NM	0.002582	2114768
10	GA	0.002542	10822590
2	AZ	0.002511	7268175
43	TX	0.002486	29640343
0	AL	0.002462	5054253
14	IN	0.002348	6811752
17	KY	0.002261	4510725
16	KS	0.002207	2937569
3	AR	0.002141	3032651
12	ID	0.002014	1893297
24	MS	0.002010	2951438
28	NV	0.001910	3141000
50	WY	0.001656	579761

The code chunk below answers the question of what states have the largest and smallest ratio of health care professionals using SQL

```
con.execute("ALTER TABLE small_healthcare ADD phys_wkforc_ratio_23 FLOAT")
con.execute("UPDATE small_healthcare SET phys_wkforc_ratio_23 = phys_wkforc_23/popn_pums_23")
display(con.execute("SELECT st_abbrev, phys_wkforc_ratio_23, popn_pums_23 FROM small_healthcare"))
```


	st_abbrev	phys_wkforc_ratio_23	popn_pums_23
0	DC	0.009578	672079
1	MA	0.004786	6992395
2	NY	0.004446	19872319
3	HI	0.003972	1445635
4	CT	0.003953	3598348
5	RI	0.003948	1095371
6	MD	0.003654	6170738
7	VT	0.003648	645254
8	PA	0.003629	12986518
9	OR	0.003590	4238714
10	OH	0.003376	11780046
11	DE	0.003355	1005873
12	CO	0.003228	5810774
13	NJ	0.003208	9267014
14	NH	0.003205	1387835
15	WA	0.003170	7740984
16	SD	0.003125	899194
17	ME	0.003124	1377400
18	MI	0.003112	10051595
19	IL	0.003068	12692653
20	US	0.003039	332387543
21	WV	0.003028	1784462
22	MN	0.002981	5713716
23	MT	0.002949	1105072
24	CA	0.002942	39242785
25	AK	0.002940	733971
26	FL	0.002936	21928881
27	TN	0.002913	6986082
28	VA	0.002912	8657499
29	NC	0.002906	10584340
30	ND	0.002879	779361
31	MO	0.002864	6168181
32	SC	0.002779	5212774
33	NE	0.002728	1965926
34	UT	0.002658	3331187
35	IA	0.002657	3195937
36	LA	0.002639	4621025
37	OK	0.002629	3995260
38	WI	0.002617	5892023
39	NM	0.002582	2114768
40	GA	0.002542	10822590

	st_abbrev	phys_wkforc_ratio_23	popn_pums_23
41	AZ	0.002511	7268175
42	TX	0.002486	29640343
43	AL	0.002462	5054253
44	IN	0.002348	6811752
45	KY	0.002261	4510725
46	KS	0.002207	2937569
47	AR	0.002141	3032651
48	ID	0.002014	1893297
49	MS	0.002010	2951438
50	NV	0.001910	3141000
51	WY	0.001656	579761

Plotting

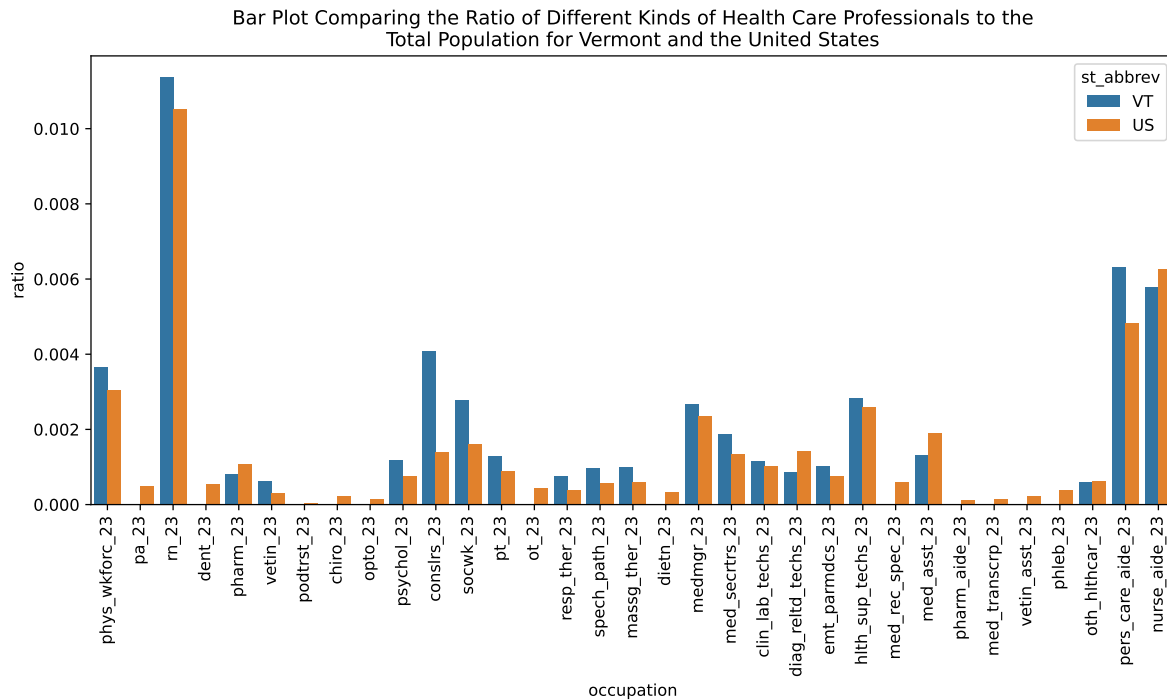
The goal of this section is to now dive into how Vermont compares to the United States, as well as other states like Vermont

The code chunk below does the plotting for the first plot

What this plot is supposed to show is the ratio of different workers in different health care professions in Vermont compared to the US as a whole. This will be helpful in understanding how Vermont's healthcare system compares to the US.

```
import matplotlib.pyplot as plt
import seaborn as sns

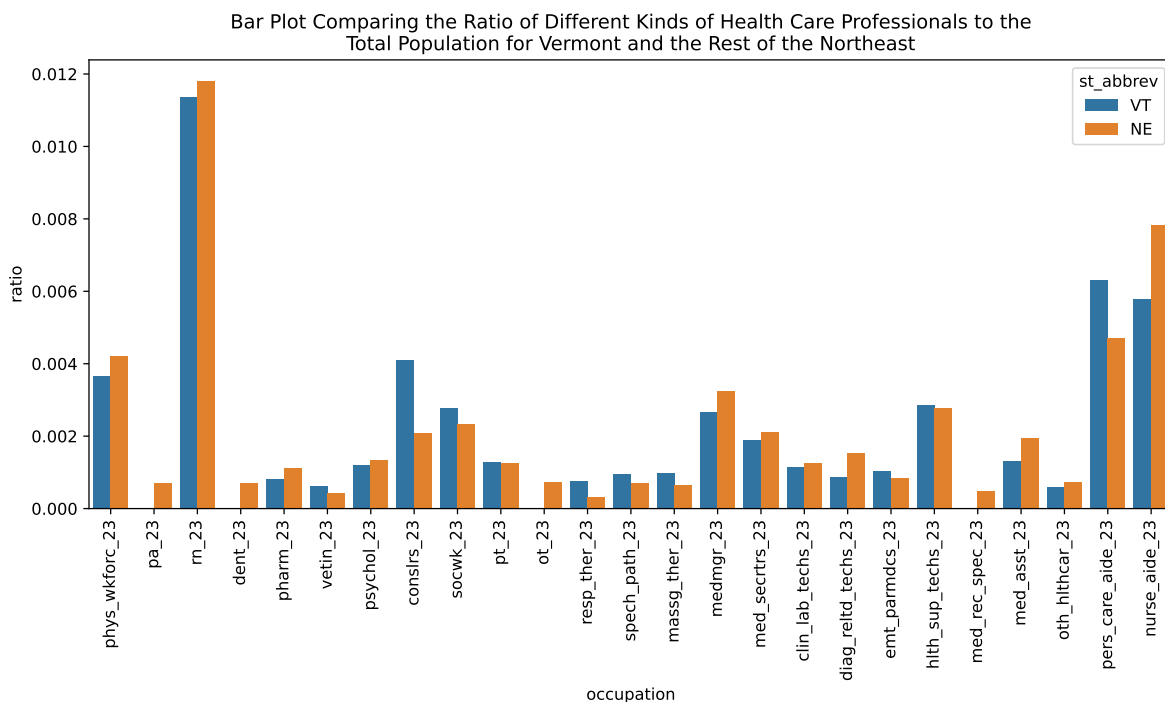
plt.figure(figsize=(12, 5))
plt.xticks(rotation=90)
sns.barplot(data=small_combined_ratio_melted, x='occupation', y='ratio', hue='st_abbrev')
plt.title("Bar Plot Comparing the Ratio of Different Kinds of Health Care Professionals to t")
plt.show()
```



Looking at the plot now, we see that Vermont has more physicians than the US on average, and tends to have a slightly higher ratio of health professionals in general. While there are a number of professions where there is no blue bar, this is because there was no data collected for those professions, rather than there being basically no one practicing them. Besides the fact that the professions with no data in Vermont are generally the least numerous professions, I know that at least some of the lack of data is because it is not collected not that there is no one practicing those professions, as one bar with no data is `ot_23`, standing for occupational therapist and I know there are occupational therapists currently practicing in Vermont.

The code chunk below does the plotting for the second plot

```
plt.figure(figsize=(12, 5))
plt.xticks(rotation=90)
sns.barplot(data=vt_and_ne_ratio_melted, x='occupation', y='ratio', hue='st_abbrev')
plt.title("Bar Plot Comparing the Ratio of Different Kinds of Health Care Professionals to t")
plt.show()
```



This plot takes looking at the whole of the United States, and narrows it down to the rest of the northeast. This means that Vermont is being compared to similar states, namely Connecticut, Maine, Massachusetts, New Hampshire, and Rhode Island. This was done by adding the the profession numbers as well as the populations of those states together, and dividing the profession numbers by the summed population to get a ratio of professionals to population for the entire Northeast leaving out Vermont. Vermont is left out because there are not too a lot of states in the Northeast, and if Vermont was left in, its numbers would skew the Northeast average closer to Vermont's, and the goal here is not to find the northeast average, but to compare Vermont's health care profession ratio's to other similar states.

Looking at the results, we can see that Vermont and the other Northeast States are pretty similar. For most of the ratio's, we can see that the other Northeast States have on average slightly more health-care professionals compared to their total population. That being said, Vermont does have a substantially larger ratio of counselors and personal care aides, as well as slightly more social workers, repository therapists, speech pathologists, and massage therapists.

Conclusion

In conclusion, Vermont is similar to the rest of the Northeast when it comes to the ratio of health care professionals to its total population. That being said, the Northeast as a whole has a larger ratio of health professionals to its population, and Vermont tends to have ratios greater than the United States as a whole.

If your looking for something surprising, the amount of counselors Vermont has is quite surprising, with its ratio being about double the ratio that the United States had. Beyond that, there was nothing too surprising in the data to me as a non-expert, though if you are an expert some of the numbers might be more shocking.

While I am wrapping up this project right now, if I have more time to devote to this then doing analysis on the cosine similarity of and k-means clustering on the ratio's of health professions between states would be interesting. I picked the Northeast region as a whole when looking for other states that are similar to Vermont, but it would be interesting seeing what the data actually says about which states are similar.